LAB TASK 2

Question 1: (Find Duplicates).

CODE	Explanation
#include<iostream> using namespace std; class Dupli{ private: int* arr; int n; int count; int maxCount; public: Dupli() : count(0), maxCount(0) , n(12) { arr = new int[n]; cout<<"Enter the values in array"<<endl; for(int i = 0; i < n; i++) cin >> *(arr + i); } void sortArr() { for(int i = 0; i< n ; i++) { for(int j = 0; j < n; j++) { if(*(arr + i) < *(arr + j)) { int temp = *(arr+i); *(arr+i) = *(arr+j); *(arr+j) = temp; } } } } void findDuplicates() { for(int i = 0; i < n; i++) { if(*(arr + i) == *(arr + i + 1)) count++; else count = 0; if(count > maxCount) maxCount = count; } } void display() { count = 1;	➢ Make a class **“Dupli”.** ➢ Create an array dynamically. ➢ Sort the array for the ease to find duplicates. ➢ Make a variable **count and initialize it with 0.** ➢ If element is equal to the next element then increase number of counts. ➢ Make another variable **Maxcount. If count is greater than maxCount** then initialize max count to count. ➢ Now maxcount can store maximum number of duplicates. ➢ Print the most duplicate number using loop.

`` for(int i = 0; i < n; i++) { if(*(arr + i) == *(arr + i + 1)) count++; else count = 0; if(count == maxCount) { cout<< "The most repeated value is: "<<arr[i] <<" and it appears "<< maxCount + 1<<" times"<<endl; } } } // void dis() { // for(int i = 0; i < n;i++) // cout<<arr[i] <<" "; // } }; int main() { Dupli D1; D1.sortArr(); D1.findDuplicates(); // D1.dis(); D1.display(); return 0; } ```	

Question 2: (Remove Duplicates)

Code	Explanation
`` #include<iostream> using namespace std; class RemoveDupli{ private: int size; int* arr; int* newarr; int newSize; public: RemoveDupli() : size(7), newSize(0) { arr = new int[size]; newarr = new int[newSize]; cout<<"Enter the values: "; for(int i = 0; i < size ; i++) cin >> *(arr + i); } ```	➢ Make a dynamic array ➢ Input values from the user. ➢ Check if there are any duplicates. ➢ Create a new array dynamically and store the uniques values from ➢ previous array in it. ➢ Display the update new array to the user.

```
    void sortArr() {
        for(int i = 0; i < size; i++) {
            for(int j = i+1; j < size; j++) {
                if(*(arr+i) > *(arr+j)) {
                    int temp = *(arr+i);
                    *(arr+i) = *(arr+j);
                    *(arr+j) = temp;
                }
            }
        }
    }

    void removeDupli() {
        for(int i = 0; i < size; i++) {
            if (i == 0) *(newarr + newSize++)
= *(arr +i) ;
            else if (*(arr + i) != *(arr+ i-1))
*(newarr + newSize++) = *(arr+ i);
        }
    }
    void display() {
        cout << "Updated sorted Array: ";
        for(int i = 0; i < newSize; i++) {
            cout << *(newarr+ i);
            if(i != newSize-1) cout << " ";
        }
    }
};

int main() {
    RemoveDupli R1;
    R1.sortArr();
    R1.removeDupli();
    R1.display();

    return 0;
}
```

Question 3: (Sum Of Diagonals).

<table>
<tr><th>Code</th><th>Explanation</th></tr>
<tr><td>

```
#include<iostream>
using namespace std;
class DiagonalSum{
private:
  int** arr;
  int mainSum;
  int SecSum;
  int size;
public:
  DiagonalSum() : size(3) , mainSum(0),
SecSum(0) {
    arr = new int*[size];
    for(int i = 0 ; i < size ; i++) *(arr + i) = new
int[size];

    cout<<"Enter the values";
    for(int i = 0 ; i < size ; i++) {
      int *a = *(arr + i);
      for(int j = 0 ; j< size ; j++) {
        cin >> *(a + j);
      }
    }
  }
  void sums() {
    mainSum = 0;
    SecSum = 0;
    for(int i = 0; i < size; i++) {
      int *a = *(arr + i);
      for(int j = 0; j < size; j++) {
        if(i == j) mainSum += *(a + j);
        if( i+ j == 2) SecSum += *(a + j);
      }
    }
  }
  void display() {
    cout<<"Main diagonal sum is: "<<
mainSum<<endl;
    cout<<"Secondary diagonal sum:
"<<SecSum<<endl;
  }
```

</td><td>

- Create a dynamic 2-D array.
- Input values from the user.
- Check if index (I == j) if **yes** then find the sum of those values and store in a variable.
- Check if sum of I and j equals to 2. If yes then calculated sum of those values and store in another variable.
- Display both values to user

</td></tr>
</table>

}; int main() { DiagonalSum S1; S1.sums(); S1.display(); return 0; }	

DATED: 2G/08/2025